Madhesh S

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PROFESSIONAL SUMMARY

B.Tech Information Technology student with practical exposure through internships, hackathons, and hands-on projects in **AI**, **ML**, and **Full Stack Development**.

EDUCATION

Saranathan College of Engineering 2027
B.Tech Information Technology *CGPA : 7.8/10*

Montfort School 2023
Class XII *Percentage : 77%*

Montfort School 2021
Class X *Percentage : 91%*

WORK EXPERIENCES

Sustainability Intern | Internship Dec 2025 - Jan 2026
AICTE | IBM SkillsBuild *online*

- Built an **AI-based food waste prediction system** using XGBoost to forecast canteen waste quantity.
- Deployed an interactive **Streamlit dashboard** with real-time predictions and Plotly-based historical data visualization.
- Generated synthetic training datasets and validated model performance against defined test scenarios for reliability.

Technologies / Skills Used : *XGBoost, Python, Streamlit*

Full Stack Development Intern | Internship Jun 2025 - Jul 2025
VDart Academy *On-site • Trichy*

- Built an **Attendance Management System** using React frontend and Django REST Framework backend architecture.
- Implemented **JWT authentication**, employee/admin portals, clock in/out tracking, and leave approval workflows.
- Developed attendance reports with **CSV export** and a real-time dashboard for employees and administrators.

Technologies / Skills Used : *React, Django, SQLite, JWT*

PROJECTS

AI-Driven Emotion-Based Adaptive Learning Platform | Github

- Developed an adaptive learning platform using **multimodal emotion recognition** for personalized learning.
- Integrated **emotion detection** and LLMs for quizzes, explanations, recommendations, and doubt resolution.
- Learner emotions engagement, boredom, confusion, frustration drive adaptive personalized learning experiences.

Technologies / Tools Used : *React, Django, MobileNetV2, DICE Dataset, Whisper*

AI-Powered Driver Drowsiness Detection | Github

- Developed a real-time AI drowsiness detection system using **OpenCV and MediaPipe** with EAR/MAR analysis.
- Created a **Node.js backend** with WebSockets, MongoDB, and MQTT for real-time driver status updates.
- Designed an **ESP32-based safety system** that triggers a buzzer, reduces speed, and activates emergency braking.

Technologies / Tools Used : *OpenCV & MediaPipe, Node.js, MongoDB, React.js & WebSockets, ESP32 & MQTT*

Forest Emergency Communication System (FECS) | Github

- Developed a **Forest Emergency Communication System** using ESP32, LoRa, and GPS for offline connectivity.
- Implemented secure LoRa mesh networking with **AES-128 encryption** and autonomous relay nodes.
- Designed an intelligent rescue dashboard with live GPS tracking and optimized routing capabilities.

Technologies / Tools Used : *React, React-Leaflet, PostgreSQL, GPS Module, ESP32 & LoRa (SX1276/78), FastAPI*

SKILLS

Databases :	MongoDB, PostgreSQL
Frontend :	React.js, HTML, CSS, JavaScript
Programming Languages :	Java, Python, SQL, C
Core CS Concepts :	DSA, DBMS, System Design, OOP
Tools & Platforms :	GitHub, Postman, Docker

CERTIFICATIONS

- Data Structures and Algorithms using **Python** (NPTEL)
- Database Management System (NPTEL)
- Introduction to Large Language Models (NPTEL)
- Programming in **Core Java** (T4TEQ)
- SOAR AI to Aspire Skill Competency (Skill India)

AWARDS & ACHIEVEMENTS

- **1st Place**, Noob HackFest 2026 — **24-hour Global Hackathon**, awarded **₹25,000 cash prize**.
- **2nd Place**, College IoT Project Expo — Recognized for innovative IoT solution.
- **Top 5%**, NPTEL DBMS Certification and **Top 5%**, NPTEL Human-Computer Interaction Certification nationally.